

Chapter 12: Analysis of Bimodal Resonance and Polarization Alignment

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The conventional General Relativistic interpretation of a binary black hole merger restricts the post-merger signal to a rapidly decaying Quasinormal Mode (QNM) ringing at a single fundamental frequency. However, the O4b strain data from GW250114 reveals a distinct, anomalous frequency split ($\Delta f \approx 15\text{Hz}$) between the primary interferometers, with H1 peaking at 115.03 Hz and L1 isolating a resonance at 130.03 Hz.

1. The Tensor and Scalar Modes

1. The Tensor Mode h_{ij} : The standard transverse-traceless gravitational wave propagating through our 4D brane, oscillating at 115.03 Hz.
2. The Scalar Mode (φ): The extra-dimensional "breathing" mode of the interface ringing at 130.03 Hz.

2. Empirical Evidence: Chirp Mass Dynamics

Analysis of the effective chirp mass (ECM) trajectory reveals a non-linear coupling regime. H1 records a sustained energy dissipation epoch ($\text{ECM} \approx 5$ at $t = -0.16\text{s}$), while L1 records a distinct, independent harmonic event at $t = -0.12\text{s}$.

This 40ms temporal divergence ($\Delta t \approx 0.04\text{s}$) is the definitive signature of a scalar-mode phase-lag relative to the primary tensor strain, anchoring the physical dimensions of the topological interface nodes to a scale of approximately 10^{-15} m , confirming the Kaluza-Klein compactification limits derived in Appendix E.